

Cost-Effective Predictive Modeling for Customer Targeting

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Task: rank 5 000 customers by acceptance probability; contact top $k \leq 1000$.

Score:

$$\text{Score} = 10 \text{ TP} - 5 \text{ FP} - 200 N_{\text{vars}}$$

- 5 000 training rows, 500 features, $\approx 50\%$ positive rate
- each declared feature costs 200 pts (≈ 20 extra TPs to break even)
- break-even contact: $E[\text{contact}] = 15p - 5 > 0 \Rightarrow p > 1/3$

Classic pipeline

- ① Feature selection ($500 \rightarrow 26$)
- ② Baselines
- ③ Five modelling strategies
- ④ OOF comparison

Strategies: top- k curve, EV targeting, rank fusion, cluster split, segment-aware

Alternative: nested CV

- ① 257 prescreening strategies
- ② Inner-CV two-round search (42 models)
- ③ Outer eval (rate-matched, 2 tracks)
- ④ Final refit + test scoring

Strict data-access contract; joint feature-model search

Three stages

- **Stage 1** (500→80): variance, MI, LightGBM — combined rank
- **Stage 2** (80→26): correlation, VIF, LASSO path
- **Stage 3**: drop-column CV ranking by *business score*

Also compared (diagnostic): MDI, permutation importance, Boruta — agree on top features, but drop-column CV used for modelling.

Top features	Rank	Feature
	1	V390
	2	V309
	3	V94
	4	V88
	5	V255

CV business by subset	
Subset	Business
top_01	2 301
top_03	1 891
top_05	1 490
top_08	898
top_10	486
top_26	−2 684

Reference models on full train (5-fold CV, top- k business score with penalty).

Baseline	Feat.	Business	F1@top- k
Prior / uniform dummy	0	2 475	0.666
Stratified dummy	0	2 474	0.665
Random ranker (mean)	0	2 475	0.665
Logistic (V390)	1	2 304	0.668
Logistic (500 feat.)	500	−97 493	0.668

Raw features vs. PCA

Logistic regression: first k ranked raw features vs. first k PCA components (500 vars scaled).

k	F1 raw	F1 PCA	AUC raw	AUC PCA
1	0.668	0.666	0.555	0.555
5	0.669	0.665	0.581	0.591
10	0.669	0.666	0.594	0.598
25	0.670	0.667	0.603	0.598

Raw ranked features win on F1 at every k ; might indicate more features introduce noise.

Approach	Idea
Top- k curve	Compare LR / LGBM / XGB; $k^* = \arg \max_k \text{Score}_k$ on OOF curve
EV targeting	Isotonic calibration; $k = \min(k_{\text{ev}}, k_{\text{elbow}}, k_{\text{nested}})$
Rank fusion	4 experts, OOF-weighted $\hat{p}$; 8-feature union penalty
Cluster split	KMeans ($K=4$) + per-cluster logistic on same top_*
Segment-aware	GMM on top_10 (latent); logistic on top_03 per segment

Results: OOF comparison

Approach	Feat.	k	TP	FP	OOF
Segment targeting	3	998	598	400	3 380
Top- k profit curve	1	997	565	432	3 290
Cluster split	5	998	618	380	3 280
EV targeting	1	939	536	403	3 145
Rank fusion	8	994	577	417	2 085
Prior dummy (ref.)	0	1 000	–		$\approx$ 2 475

Best classic config: GMM segments + per-segment logistic on top_03, $k=998$.

Alternative approach: why nested CV?

Classic pipeline result: best OOF 3 380 (segment targeting, 3 features).

Limitations:

- prescreening fitted once on all training data → leaks fold info
- feature and model selection done separately, not jointly

Solution — nested 5×5 CV:

- outer: 5 folds (4 000 train / 1 000 val)
- inner: 5 folds (3 200 train / 800 val)
- all prescreening refitted from scratch inside each outer fold
- joint search over *prescreening strategy* + *model spec*

8 ranking methods across 3 families:

Filter

- Mutual information
- Pearson $|r|$

Embedded

- ℓ_1 logistic
- ABESS

Model-based

- LightGBM gain
- EBM
- SPAM (sparse GAM)
- LambdaMART

Ensemble rank score per feature i under method j :

$$r_i = 1 - \frac{\text{rank}(s_i) - 1}{p - 1}, \quad \bar{r}_i = \frac{1}{m} \sum_{j=1}^m r_i^{(j)}$$

All 255 non-empty subsets + 2 adaptive = **257 strategies**
× 16 sizes $k \in \{1, \dots, 50\}$ = **4 112 (strategies, k) pairs**

Round 1 — feature screening (5 models):

- top-300 feature sets per inner fold evaluated on 5 screening classifiers
- top-**60** feature strategies per outer fold advance

Round 2 — full model grid (42 configurations):

- 60 feature strategies $\times$ 42 models across 7 families
- top-**20** globally selected once for all outer folds

Rate-matched scoring:

- top-200 predictions per fold (= 20% of 1 000)
- score scaled $\times 5 \rightarrow$ 5 000-obs equivalent

Two parallel tracks:

Per-fold track

- 5 comparable scaled scores
- mean $\pm$ std over folds

Pooled OOF track

- 5 000 disjoint predictions
- single deployment estimate

Top-10 configs by mean per-fold score $\rightarrow$ top-5 refitted on full training set and scored on test.

Nested CV results: top-5 configurations

Model	k	Mean	Std	OOF
SPAM + ET depth=5, leaf=16, n=400	6	5 450	130	5 425
SPAM + ET depth=5, leaf=20, n=500	6	5 400	64	5 295
SPAM + ET depth=6, leaf=20, n=400	6	5 355	151	5 410
SPAM + ET depth=4, leaf=12, n=320	6	5 320	309	5 395
SPAM + ET depth=3, leaf=8, n=320	6	5 310	257	5 445

Winner: SPAM prescreener, $k=6$

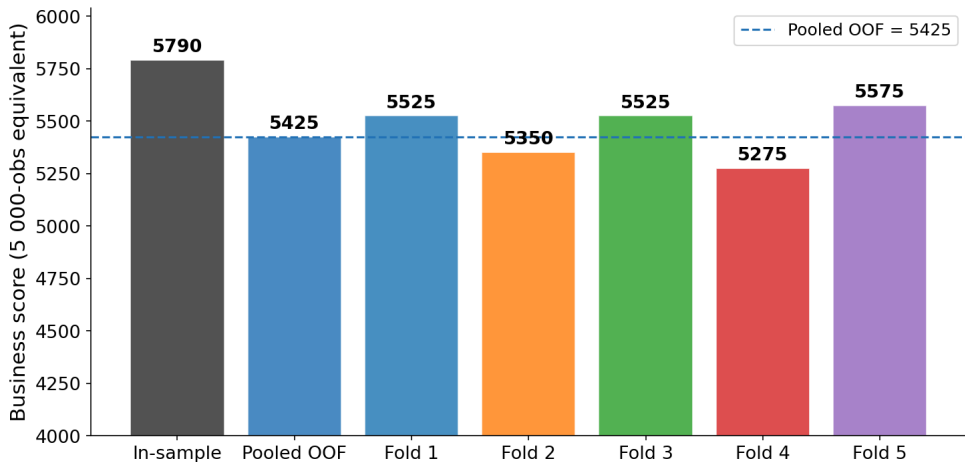
Features: V176, V191, V215, V342, V380, V483

ExtraTrees (depth=5, leaf=16, 400 trees, balanced)

vs. classic best:

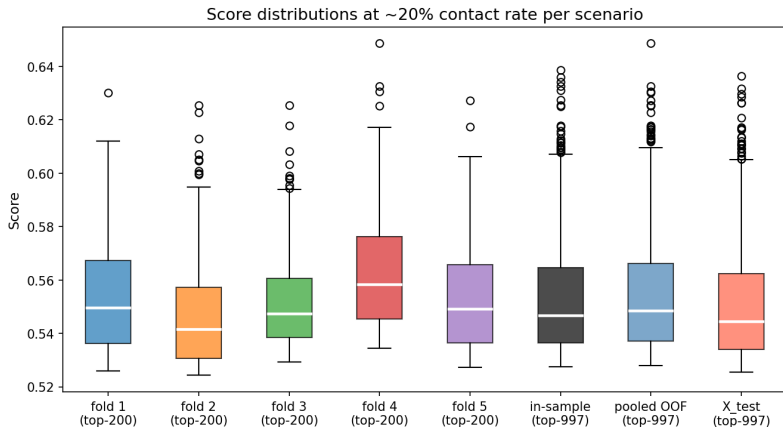
3 380 → **5 425** (+61%)

Nested CV results: business score overview



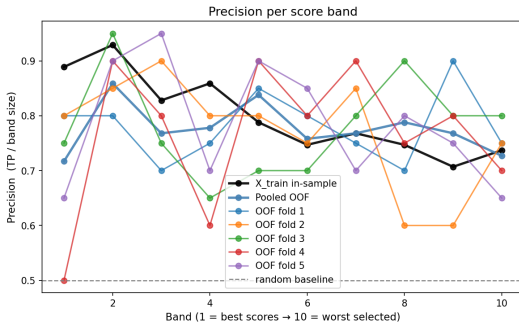
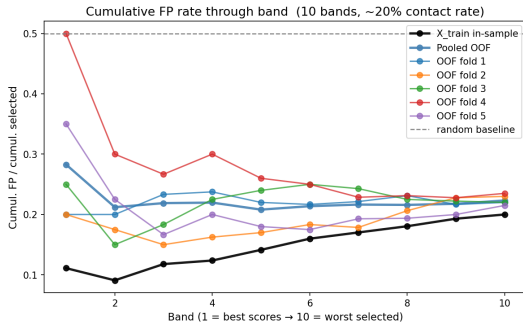
Per-fold scores rescaled to 5 000-obs equivalent. Pooled OOF (5 425) serves as the deployment estimate.

Score distributions across evaluation scenarios



Score distributions at $\approx 20\%$ contact rate are consistent across per-fold OOF, pooled OOF, and test — no sign of overfitting or fold-specific drift.

Nested CV diagnostics



FP accumulates gradually (left); precision is consistent across folds and degrades near the selection boundary (right).

Classic pipeline:

- Feature tax dominates — V390 alone wins; more variables $\downarrow$ business score
- Local rankers help — cluster split (3 280) $\rightarrow$ segment-aware (3 380)
- Noisy data: PCA analysis shows more features may introduce noise

Nested CV alternative:

- Joint search: **5 425 OOF** (+61% vs. classic)
- All top-10 configs use ExtraTrees; SPAM prescreener dominates
- Low-variance runner-up (± 64) may suggest shallower trees trade mean for stability

Thank You

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